CARA Kinematics Testing: Baseline vs Analytical

23 October 2026

Test setup

The test uses the one-leg CAD model exported as MuJoCo XML. The foot follows a base circle and then a rising spiral. Two runs are saved: a MuJoCo baseline and a run using the analytical IK angles.

Setting	Value
Trajectory center	(-80, 65, -240) mm
Radius / rise	35 mm / 60 mm
Path	One base circle, followed by four shrinking rising turns
Samples / nominal duration	1,601 / 16 s (4 s base circle and 12 s rise)
Position frame	Relative to FL_O; world-aligned axes; Z upward
Input joint order	FL_abad, FL_a, FL_b
Simulation	Contacts off; gravity off; fixed base; 1 ms timestep
Software reported by the run	Python 3.14.6; MuJoCo 3.17.0

How the test runs

1. Baseline: the foot is connected to a movable target through an equality constraint. The target moves through the trajectory points, and MuJoCo settles the linkage at each point. The measured three joint angles, foot XYZ, joint states and settling values are stored.

2. Analytical run: each desired XYZ point is passed to the analytical IK. The calculated angles are applied to the three independent joints. The foot-target constraint is disabled, and MuJoCo measures the resulting foot XYZ while keeping the linkage loop connected.

3. Comparison: after both runs are saved, the script reads their values and calculates the position, FK, Jacobian and velocity differences. The nominal 16 s describes the trajectory; each point has its own settling interval.

What is compared

FK: the measured baseline angles are passed to the analytical FK. Its predicted XYZ is compared with the baseline foot XYZ. IK: the achieved foot XYZ from the analytical angle-driven run is compared with the desired XYZ. The achieved positions from the two runs are also compared directly.

The analytical Jacobian is compared with the CAD effective Jacobian, which includes passive-joint motion through loop closure. Both Jacobians use the same joint rates, (5, 10, -7) degrees/s, to calculate foot velocity. The velocity difference is calculated from these two predictions.

Position and velocity errors use the Euclidean norm. Jacobian error is the largest absolute entry difference. RMS = sqrt(mean(error squared)).

Recorded values

Comparison	Maximum	RMS
Baseline to target (mm)	3.62463e-05	1.58052e-05
Analytical FK to baseline (mm)	1.21541e-05	4.74573e-06
Analytical IK run to target (mm)	6.16732e-05	1.58746e-05
Between the two runs (mm)	6.16718e-05	1.54151e-06
Jacobian entry (mm/rad)	1.11358e-05	4.25088e-06
Predicted velocity (mm/s)	1.07938e-06	3.80283e-07

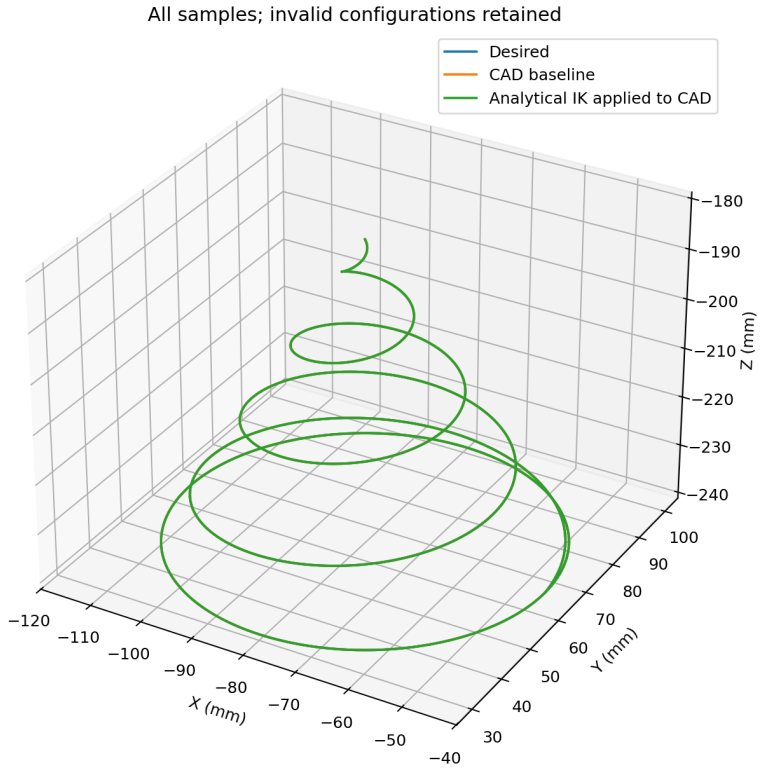
All 1,601 baseline samples and all 1,601 analytical-run samples meet the configured pose checks. The comparison contains 1,601 valid pairs; failures.csv has no failed samples.

Settling value	Baseline maximum	Analytical maximum
Loop closure gap (mm)	1.61760e-5	3.05941e-5
Joint speed (rad/s)	5.05252e-5	8.48350e-5
Relaxation steps per point	50	68

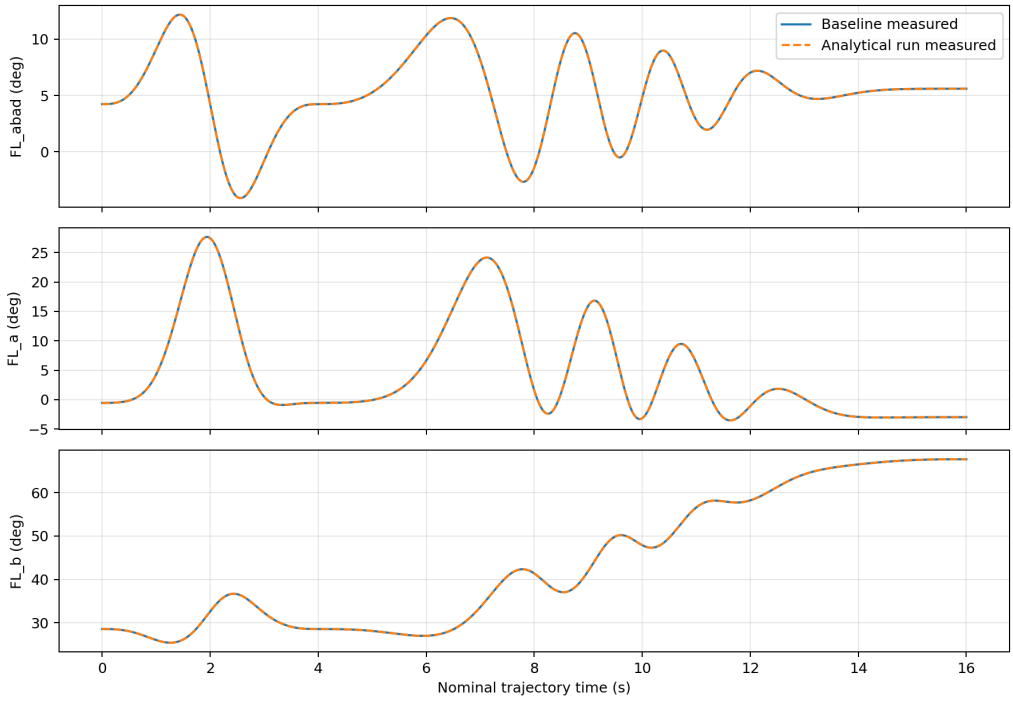
The maximum analytical angle-holding error is 2.07563e-5 degrees. The largest IK target error and the largest difference between runs occur at the first sample. After the first sample, the maximum difference between the two achieved paths is 6.70356e-8 mm.

The configured comparison limits are 0.1 mm for FK and IK, 0.01 mm/rad for the Jacobian, and 0.01 mm/s for predicted velocity. All four checks are marked as passing in summary.json.

Trajectory and joint plots



Desired path, MuJoCo baseline and analytical IK-driven path are plotted together. Their curves overlap at this scale.



Measured joint angles from both runs are plotted against nominal trajectory time.

Saved files

Each run stores a sample CSV, full joint states, a simulation recording and a trajectory playback. The comparison stores the error CSV, summary, failures table and plots. This report uses the supplied CSV, summary and figures; no additional simulation was run.